Gate Before You Rank: Soundness Buys a Floor, Not a Gain, in Verifier-Based Test-Time Scaling

Anonymous ARR submission

DRAFT STATUS: THEORY COMPLETE, EXPERIMENTS NOT RUN. Sections 4 and 5 are finished and the propositions were checked numerically. Every entry in Tables 1 and 2 is an unmeasured placeholder and §7 states predictions, not findings. Figure 1c plots a closed-form consequence of Cor. 2 and contains no measured data. The table in Appendix F reports figures taken from other papers, not measured here. Do not submit until §6 has been run. See `BLOCKERS.txt`.

Abstract

Verifier-based test-time scaling is bounded by verifier quality, and Dorner et al. [2] show the bound is governed by the geometry of the verifier's ROC curve. Best-of-N ranks a pool that contains the base output, so it can discard a correct base answer however good the verifier is. One change removes that risk: *gate* on the base output first and rank only after the gate has rejected it. Paired with a verifier at the *sound* corner of the curve, where a correct output is never rejected, the gate turns verifier quality into an unconditional guarantee. We prove that accuracy then cannot fall below the base at any budget, for any scoring function, with no assumption on how candidates are generated; that soundness is closed under conjunction, so cheap checks stack and push the false-acceptance rate ε down without endangering the floor; and that any scorer inherits the floor when gated behind a sound check, so an outcome reward model belongs *behind* a structural check rather than in competition with one. What the gate does not buy is accuracy. The gain factors into how often the base is wrong, how often the verifier catches it, and how good the replacement is, and soundness appears in none of the three: it sets the floor and leaves every unit of improvement to the false-acceptance rate ε. That predicts cheap structural checks are safe but weak, and would explain why execution-based best-of-N loses to learned reward models on text-to-SQL [13]. The floor also carries a liability: soundness is an argument about a task but is realised by an implementation, and implementations leak, so we separate *nominal* from *realised* soundness, price the gap, and make auditing it the first experiment, not an assumption. **[RP: measured $p_0, p, \varepsilon, \delta$ across three structured-generation tasks, and observed versus predicted gain]**

1 Introduction

Sampling several candidate outputs and selecting among them improves accuracy on reasoning and code tasks [1, 11, 14], and the ceiling on that improvement is set by the verifier rather than by the compute budget. Stroebl et al. [12] make the point sharply: if a verifier accepts an incorrect candidate with any fixed probability, resampling cannot drive that probability down, so accuracy saturates. Dorner et al. [2] give the general picture, proving that the accuracy of rejection sampling and best-of-N is determined by the geometry of the verifier's ROC curve. Learned verifiers, whether reward models or process supervision [5], sit somewhere in the interior of that curve.

Structured generation looks like an escape route. When a model emits SQL, a typed program, a JSON record, or a morphological segmentation, the output space carries hard constraints that a correct answer satisfies by construction. A gold SQL query parses, references only existing columns, and executes. A gold segmentation concatenates back to its surface form. Checking these costs microseconds, needs no labels, and needs no training. If the bottleneck on test-time scaling is verifier quality, and structured NLP hands us verifiers that never reject a correct answer, the bottleneck should lift.

This paper asks what that corner of the ROC curve is actually worth. The answer has a good half and a bad half, and the bad half is not the one the field has been discussing.

The two rates. Write $\delta = \Pr[\text{reject} \mid \text{correct}]$ for the false-rejection rate and $\varepsilon = \Pr[\text{accept} \mid \text{incorrect}]$ for the false-acceptance rate. In ROC terms a sound verifier is the point with true-positive rate 1: it lies on the top edge, at whatever ε its constraint happens to induce. Soundness pins one coordinate and says nothing about the other. A syntactically valid SQL query that executes cleanly and returns the wrong rows passes every structural check there is, so we expect ε near one and a nearly vacuous filter.

Nominal versus realised soundness. Soundness is established by an argument about the task: a gold query executes, so executability cannot reject a correct answer. But the argument is about the task and the check is an implementation, and implementations leak. A correct answer can be rejected because the extractor failed to find the SQL block, because the normaliser did not recognise an equivalent formulation, or because the output was embedded in explanatory prose. This is not hypothetical. Among responses a rule-based math verifier marked incorrect, 38.5% were in fact correct [6], and rule-based checkers are systematically brittle to formatting and algebraic equivalence [8]. We therefore distinguish *nominal* soundness, the property the task argument establishes, from *realised* soundness, the measured δ of the deployed check. The floor below holds for the latter, not the former, and the gap between them is where it is lost.

What this predicts, and what it explains. §4 makes this exact. Under a reject-gated rule that consults candi-

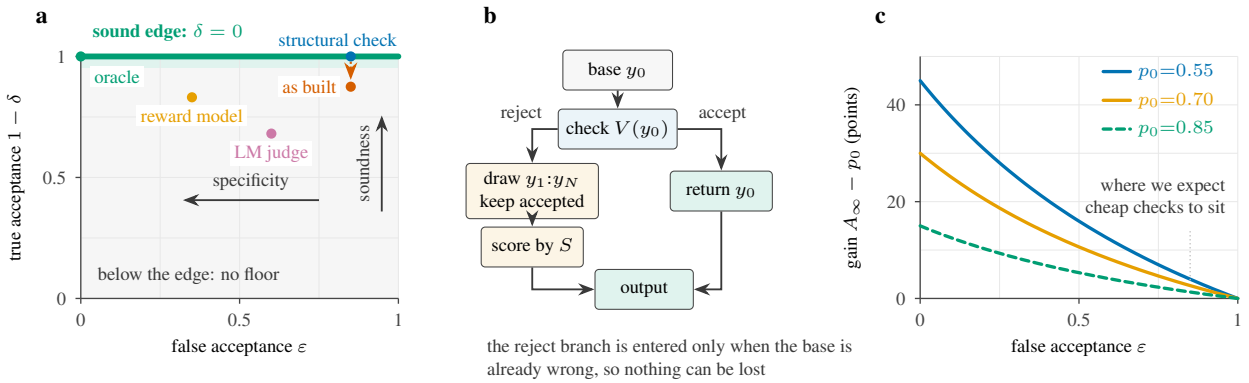


Figure 1: Verifier quality is two numbers, and soundness pins the one that does not pay. (a) Any verifier is a point in the plane of its two error rates. Soundness is the top edge, $\delta = 0$, where a correct output is never rejected; only on that edge does Prop. 1 hold, and everywhere below it the floor is forfeit. Moving *up* buys the guarantee, moving *left* buys accuracy, and a structural check is a point that is already at the top and far to the right. *Marker positions are illustrative, not measured*: placing them is what Experiment 2 does, and the paper's central prediction is that the blue marker sits where it is drawn. **(b)** The reject-gated rule. Because a sound rejection certifies that the base is wrong, the sampling branch is reached only on instances already lost, which is why the floor needs no assumption about the scorer, the budget, or the candidate distribution. **(c)** What the edge is worth, from the closed form in Cor. 2. Gains are in accuracy points throughout, matching Tables 1 and 2: the gain collapses toward zero well before ε reaches one, and collapses faster for a stronger base model. **Curves are a formula, not data.**

dates only when the base output is rejected, accuracy is

$$A_N \;=\; p_0 \;+\; \underbrace{(1-p_0)}_{\text{headroom}}\;\underbrace{(1-\varepsilon)}_{\text{caught}}\;\underbrace{\rho_N}_{\text{repl.}},$$

where $q = p + (1-p)\varepsilon$ is the probability that a sampled candidate is accepted, $r = p/q$ is the chance an accepted candidate is correct, and $\rho_N = \big(1-(1-q)^N\big)r$ discounts r by the chance that no candidate is accepted at all. Section 4 keeps q and r and defines them for general δ; ρ_N is a convenience used only here. Three consequences follow. Accuracy never falls below p_0, for any N, any task, any ε, provided δ is genuinely zero: realised soundness buys a floor. The gain is proportional to $(1-\varepsilon)$ and vanishes as $\varepsilon \to 1$: soundness buys no accuracy on its own. And a verifier with $\delta > 0$ can fall below p_0, under a condition we state exactly, which is what makes the nominal-versus-realised gap costly and not merely untidy.

The middle consequence would resolve a result that otherwise looks like a puzzle. Tritto et al. [13] report that learned outcome reward models beat execution-based best-of-N on BIRD and Spider, which is surprising if execution is the sound verifier that soundness arguments recommend. On our account it is expected: execution has $\delta \approx 0$ but ε near one, so it contributes a floor and little else, while a reward model trades a positive δ for a much smaller ε and wins on net. Appendix F collects what the literature already reports about both rates. Whether this explanation is right is an empirical question about ε, and Experiment 2 settles it.

Contributions.

- **An analysis of the sound corner of the ROC curve** (Prop. 1 and Thm. 1). Dorner et al. [2] characterise accuracy across the whole curve for rejection sampling and best-of-N; neither method has a base-gating step, and neither singles out the top edge. We show that adding the gate makes the $\delta = 0$ point qualitatively different from its neighbours: it acquires an unconditional floor that survives any scorer and any ε. The propositions are elementary and each is verified numerically against simulation.
- **Sound verifier stacking** (Prop. 2). Soundness is closed under conjunction and ε is non-increasing under it, so cheap checks accumulate to lower ε with no risk to the floor. This turns a binary property into a design axis. It is also the operation that distinguishes our composition from verifier ensembling: Saad-Falcon et al. [9] combine weak verifiers by a learned weighted score, and any positive weight on an unsound member destroys $\delta = 0$, whereas conjunction preserves it exactly and needs no supervision.
- **Safe composition with a learned verifier** (Cor. 1). Gating an arbitrary scorer, learned or unsound, behind a sound check on the base output preserves the floor exactly. Neither the sound-verifier nor the reward-model literature states this, and it is what practitioners should deploy.
- **The soundness audit as a first-class experiment** (§6). Because nominal soundness does not imply realised soundness, we measure δ for every check claimed sound before anything else, instead of consigning the question to a footnote, and Cor. 3 prices the failure. The design then tests predicted against observed gain and includes a negative control that predicts exactly zero effect.

We claim none of structural verification, best-of-N, execution-guided selection, or the ROC characterisation of verifier imperfection; all are established, and §2 places each. The claim is narrower: that the $\delta = 0$ edge behaves differently from the rest of the curve once a gate is added, that the difference is a floor rather than a gain, and that the floor is routinely forfeited by implementations nobody

audits.

2 Related Work

Verifier-bounded test-time scaling. Stroebl et al. [12] show that a nonzero false-acceptance rate upper-bounds resampling irrespective of budget, using unit-test coverage on HumanEval and MBPP. Setlur et al. [10] argue that verifier-based training and search dominate verifier-free alternatives at matched budget, which sharpens the dependence on verifier quality rather than relaxing it. The closest work is Dorner et al. [2], who prove that the instance-level accuracy of rejection sampling and best-of-N is characterised by the geometry of the verifier's ROC curve, that rejection sampling dominates best-of-N at fixed compute while both converge in the infinite-compute limit, and that low-compute behaviour does not in general extrapolate. Their result is more general than ours and we do not restate it. We are explicit about the relationship. Our gated rule is rejection sampling in which the *first draw is the greedy output*, so it is not a new algorithm; what differs is that the first draw comes from a different distribution than the rest, with accuracy $p_0 \neq p$. Their analysis assumes draws from a single policy, and the floor in Prop. 1 is stated relative to p_0, which is the quantity that asymmetry introduces. We note explicitly that homogeneous rejection sampling with a sound verifier already cannot fall below the *sampled* accuracy p, since $p/q \geq p$ whenever $\varepsilon \leq 1$; what the gate adds is a floor at p_0 instead, and the two differ exactly when greedy decoding beats sampling. Ungated best-of-N has no floor of either kind, because it scores the base alongside the candidates and can discard it.

Both error directions, measured. That verifiers fail in two directions is established in the reinforcement-learning-from-verifiable-rewards literature [8]. [8] formalise verifier unreliability as a reward channel with asymmetric noise rates and derive corrections for it. [6] analyse the false-negative direction directly on a mathematics dataset, attributing it to formatting, notation and equivalence mismatches, and report that 38.5% of the responses a rule-based verifier rejected were in fact correct. In the other direction, language-model judges accept substantively empty responses at high rates given a superficial prefix [7]. That work asks which direction matters for *training* a policy. We ask which matters for *selecting* an output, and the answer differs: for training, false positives dominate because they are what a policy learns to exploit; for gated selection, false negatives are the only direction that can cost accuracy at all.

Selection in structured generation. Best-of-N over sampled programs, ranked by execution success or output frequency, is standard in text-to-SQL. Tritto et al. [13] train outcome reward models from execution-labelled candidates and report that they outperform both execution-based best-of-N and majority voting on BIRD and Spider. Self-consistency [14] is the frequency-based special case. Li et al. [4] attack the same gap without training, by cross-checking candidate execution against a parallel imperative implementation on a constructed database, and report gains over both. Their check is a strong *discriminator* and is not sound in our sense, since a correct query can disagree with an imperfect reference implementation; on our account that is the trade it makes, and Cor. 1 says it belongs behind a gate rather than in one. We take the reported ordering as the phenomenon to explain rather than a baseline to beat, and §6 tests whether measured ε accounts for it.

Combining verifiers. Saad-Falcon et al. [9] close part of the generation-verification gap by ensembling weak verifiers with weak supervision to estimate each one's accuracy, finding weighted combinations clearly better than unweighted ones. Prop. 2 concerns a different operation on a different class. Weighted ensembling optimises a score and cannot preserve $\delta = 0$, because any positive weight on a member with $\delta > 0$ can outvote a sound member; conjunction preserves it by construction, requires no estimation, and is available with zero labelled data. The two are complementary rather than competing, and Cor. 1 is how they combine.

Selective prediction and cascades. The reject-gated rule is structurally a cascade: score the cheap output, fall back or escalate when the score is poor. That literature runs from recent abstention work back to classical selective classification. **[MISSING: cite Chow (1957 or 1970), El-Yaniv and Wiener (2010), Geifman and El-Yaniv (2017), and one recent LLM cascade paper; the placeholder citation in this sentence is wrong and must be replaced]** The distinguishing property here is what the gate is. Cascades and selective predictors gate on a calibrated confidence, so they need threshold tuning and inherit any miscalibration; a sound rejection is a logical certificate of error, so the gate needs no calibration, no threshold, and no distributional assumption, which is why Prop. 1 carries no conditions.

Constrained decoding. Grammar-constrained and finite-state decoding guarantee well-formedness in a single pass, which raises the obvious objection that a sound filter is redundant. It is not, for two reasons. Constrained decoding requires logit access and a prefix-decidable constraint, whereas verification requires only the ability to check a finished output. Some of the constraints we care about are not prefix-decidable: whether a query executes without error against a live database depends on the completed query and on runtime state, and cannot be enforced token by token. Others plainly are prefix-decidable, and we do not claim otherwise. Concatenation of segments to a known surface form, for instance, is a simple prefix constraint that a decoder can enforce directly, and where a constraint is decodable, enforcing it is the better engineering choice. We conjecture, but do not show, that constraints with low ε tend to be the undecodable ones; that is a question the ε measurements in §6 can begin to answer and we do not rest any claim on it. §6 includes constrained decoding as a baseline. **[MISSING: cite two or three current grammar-constrained decoding**

papers and confirm this characterisation against them; do not assert it unsupported]

Anatomy of the gap. Analyses of when scaling pays off usually study the generator, through coverage or pass@k curves [1]. We hold the generator fixed and vary the verifier.

3 Setup

Fix a task instance. A model produces a base output y_0 by greedy decoding and, on demand, candidates $y_1, \ldots, y_N$ sampled i.i.d. from the same model at fixed temperature. Let $C(y) \in \{0, 1\}$ indicate correctness against the gold answer under the task metric; C is available only at evaluation time and never to the method. Write

$$p_0 = \Pr[C(y_0) = 1], \qquad p = \Pr[C(y_i) = 1],$$

noting $p_0 \neq p$ in general because greedy and sampled decoding differ.

A *verifier* is a function $V : \mathcal{Y} \to \{0, 1\}$ computable from a candidate alone, with no labels and no reference. Its two error rates are

$$\delta = \Pr[V(y) = 0 \mid C(y) = 1],$$
$$\varepsilon = \Pr[V(y) = 1 \mid C(y) = 0].$$

Definition 1 (Nominal and realised soundness)**.** *A check is* nominally sound *for a task if the task definition entails that a gold output satisfies it. It is* realised sound *on a deployment distribution if its measured δ is zero, that is, $C(y) = 1 \Rightarrow V(y) = 1$ holds for the check as implemented.*

Nominal soundness is established by argument. Executability is nominally sound for text-to-SQL because a gold query executes; concatenation is nominally sound for morphological segmentation because gold segments concatenate. Realised soundness is a measurement, and the two come apart whenever the implementation adds steps the argument did not cover: extracting the query from surrounding prose, normalising an equivalent formulation, tolerating whitespace. Exactly this failure is documented for rule-based math verifiers [6, 8]. Every result in §4 requires realised soundness. None of them follows from nominal soundness alone, and Experiment 2 is the audit that separates the two.

Nothing about either notion constrains ε, which must be measured independently, and published figures show the two rates sitting very far apart in the same systems (Appendix F).

Throughout we assume δ and ε are the same for the base output and for sampled candidates. **[VERIFY: this is an idealisation: greedy outputs may be better formed than sampled ones, which would give the base a lower ε. Experiment 2 measures both separately and the assumption is discharged or reported violated]**

Reject-gated selection. The rule we analyse is: return y_0 if $V(y_0) = 1$; otherwise draw $y_1, \ldots, y_N$, and return the accepted candidate maximising a scoring function S (uniform if S is constant), or y_0 if none is accepted. Figure 1b shows it. The gate is what makes the analysis unconditional, and it differs from the usual best-of-N rule, which scores the whole pool including the base and can therefore replace a correct base output.

4 What the Sound Corner Buys

We separate results by what they assume. Proposition 1 and its consequences need only realised soundness and hold instance by instance. The exact accuracy formula needs a model of how candidates and verdicts are generated, which we state as A1–A3 and then partly remove in §4.3. Full proofs are in Appendix A; the auxiliary lemma they use is Appendix B; the numerical check of every statement is Appendix E.

Definition 2 (Reject-gated selection)**.** *Given a verifier V, a scoring function S, and a budget N: return y_0 if $V(y_0) = 1$; otherwise draw $y_1, \ldots, y_N$ and return $\arg\max_{i \in \mathcal{A}} S(y_i)$ where $\mathcal{A} = \{i : V(y_i) = 1\}$, breaking ties uniformly, or y_0 if $\mathcal{A} = \emptyset$.*

4.1 The guarantee

Proposition 1 (Instance-wise floor)**.** *Fix any instance x and let V be realised sound at x (Definition 1), meaning $C(y) = 1 \Rightarrow V(y) = 1$ for outputs at x. Under Definition 2,*

$$\Pr\big[C(\hat{y}) = 1 \;\big|\; x\big] \;\geq\; \Pr\big[C(y_0) = 1 \;\big|\; x\big]$$

for every budget $N \geq 0$, every scoring function S, every false-acceptance rate, and every candidate distribution. Averaging over x gives $\Pr[C(\hat{y}) = 1] \geq p_0$.

Sketch. Realised soundness contraposed says $V(y_0) = 0 \Rightarrow C(y_0) = 0$. The gate therefore enters the sampling branch only on instances where the base output is already incorrect, so no correct base output is ever replaced and the event $\{C(y_0) = 1\}$ is contained in the event $\{C(\hat{y}) = 1\}$. Appendix A. □

Three features of this statement matter. It is conditional on the instance, not an average, so the guarantee is not an artifact of aggregation and cannot be paid for by a subpopulation. It assumes nothing about how candidates are generated, whether they are independent, or how S behaves. And it is conditional on exactly one measurable quantity, which is why §6 audits that quantity first.

The contrast with ungated best-of-N is what makes the corner special: ungated selection scores the whole pool including the base and can return a candidate when the base was already correct, so no amount of soundness protects it. The ROC characterisation of Dorner et al. [2] applies across the whole curve precisely because it does not assume a gate.

Corollary 1 (Safe composition)**.** *Under the hypotheses of Prop. 1, the floor holds for* any *scoring function S: learned or heuristic, calibrated or not, sound or not.*

Corollary 1 is the deployment recommendation. An outcome reward model used as an unguarded best-of-N scorer has no floor and can fall below greedy decoding; the same model, consulted only after a sound check has certified the base incorrect, cannot. The two literatures compose, and the composition dominates either alone.

Proposition 2 (Stacking)**.** *Let $V_1, \dots, V_k$ each be realised sound at x, with false-acceptance rates $\varepsilon_1, \dots, \varepsilon_k$, and let $V = \bigwedge_i V_i$. Then V is realised sound at x, and $\varepsilon_V \le \min_i \varepsilon_i$. Both statements hold with no independence assumption.*

Sketch. A correct output satisfies every conjunct, so it satisfies the conjunction and $\delta_V = 0$. Acceptance by V implies acceptance by each V_j, so the accepting event is contained in each, giving $\varepsilon_V \le \varepsilon_j$ for every j. Appendix A. □

Proposition 2 turns soundness from a property a verifier has or lacks into a design budget: cheap checks accumulate, each weakly lowering ε, with a guarantee that nothing is risked. Two cautions. The proven bound is $\min_i \varepsilon_i$; the much stronger product $\prod_i \varepsilon_i$ requires the conjuncts to fail independently on incorrect outputs, which is implausible for correlated checks such as schema validity and executability, and we never report it. And conjunction preserves realised soundness only if every conjunct has it, so k checks mean k audits and one leaky conjunct forfeits the floor for the stack.

This is also what separates our composition from verifier ensembling. Saad-Falcon et al. [9] combine weak verifiers into a learned weighted score; any positive weight on a conjunct with $\delta > 0$ can outvote a sound one, so weighted ensembling cannot preserve $\delta = 0$. Conjunction preserves it by construction and needs no labelled data.

4.2 The quantity

Quantifying the gain, unlike guaranteeing the floor, requires a model.

Assumption 1.

A1 *$y_1, \dots, y_N$ are i.i.d. with $\Pr[C(y_i)=1] = p$, and $\Pr[C(y_0)=1] = p_0$, with $p_0 \neq p$ permitted.*

A2 *Given correctness, verdicts are independent across candidates with $\Pr[V=0 \mid C=1] = \delta$ and $\Pr[V=1 \mid C=0] = \varepsilon$, identical for y_0 and the samples.*

A3 *$p_0, p, \delta, \varepsilon$ do not vary across instances.*

Theorem 1 (Accuracy of reject-gated selection)**.** *Under Definition 2, Assumption 1, and constant S, write $q = p(1-\delta) + (1-p)\varepsilon$ for the acceptance probability of a sampled candidate and $r = p(1-\delta)/q$ for the chance that an accepted one is correct. Both generalise the quantities named in §1 and reduce to them at $\delta = 0$. Assume $q > 0$; if $q = 0$ no candidate is ever accepted and $A_N = p_0$ for every N. Then*

$$A_N = p_0(1-\delta) + p_0\delta\,(1-q)^N + \big[p_0\delta + (1-p_0)(1-\varepsilon)\big]\big(1-(1-q)^N\big)r.$$

Sketch. Split on the gate. The gate passes and the base is correct with probability $p_0(1-\delta)$. The gate fails with probability $p_0\delta + (1-p_0)(1-\varepsilon)$; the sampling branch then yields a correct output with probability $(1-(1-q)^N)r$ by Lemma 1, and otherwise returns y_0, which is correct only in the $p_0\delta$ sub-case. Appendix A. □

Setting $N = 0$ returns $A_0 = p_0$ for every parameter setting, which is the identity the formula must satisfy.

Corollary 2 (Gain decomposition, sound case)**.** *If $\delta = 0$ then $q = p + (1-p)\varepsilon$ and*

$$A_N = p_0 + \underbrace{(1-p_0)}_{\textit{headroom}}\underbrace{(1-\varepsilon)}_{\textit{caught}}\underbrace{\big(1-(1-q)^N\big)\tfrac{p}{q}}_{\textit{replacement}},$$

which is non-decreasing in N and satisfies $A_\infty - p_0 = (1-p_0)(1-\varepsilon)p/q$, an expression strictly decreasing in ε.[1]

Read the three factors. $(1-p_0)$ is the headroom, the mass the base gets wrong. $(1-\varepsilon)$ is the fraction of that mass the verifier catches. p/q is the chance the replacement is right. Soundness appears in none of them; it appears only in Prop. 1. As $\varepsilon \to 1$ the gain vanishes at every budget: a check that almost every wrong answer passes cannot help, however cheap and however sound. Figure 1c plots the closed form.

Corollary 3 (Degradation)**.** *Under Assumption 1 with $\delta > 0$,*

$$A_\infty - p_0 = -p_0\delta(1-r) + (1-p_0)(1-\varepsilon)\,r,$$

so accuracy falls below the base exactly when $p_0\delta(1-r) > (1-p_0)(1-\varepsilon)r$. Setting $\delta = 0$ annihilates the first term and recovers Cor. 2.

The condition reads: a verifier degrades when the correct base outputs it wrongly discards, weighted by the chance their replacement is wrong, outweigh the incorrect base outputs it correctly catches, weighted by the chance their replacement is right. Two cases deserve naming. Confidence-based selection is dangerous because a model that is confidently wrong carries a high δ and a high ε at once. More quietly, a nominally sound check with a leaky implementation is dangerous for the same reason at smaller scale, and its users believe Prop. 1 protects them. The loss is first-order in δ: nothing has to be badly broken.

[1] At finite N the two ε-dependent factors move in opposite directions, since raising ε lowers $(1-\varepsilon)p/q$ but raises the acceptance probability q and hence $1-(1-q)^N$. The product is non-increasing in ε everywhere on the grid of Appendix E, but we have not proved it and do not assert it.

4.3 Dropping the homogeneity assumption

A3 is false in practice, since difficulty varies by instance. It can be removed from the gain accounting exactly, at the cost of one covariance term.

Proposition 3 (Heterogeneous gain)**.** *Let A1–A2 hold conditionally on each instance with rates $p_0(x), p(x), \varepsilon(x)$ and $\delta(x) = 0$, and drop A3. Write $h(x) = (1 - \varepsilon(x))\, p(x)/q(x)$ for the per-instance asymptotic replacement term. Then*

$$A_\infty - \bar{p}_0 \;=\; (1 - \bar{p}_0)\, \mathbb{E}[h] \;+\; \mathrm{Cov}\big(1 - p_0(x),\, h(x)\big),$$

where $\bar{p}_0 = \mathbb{E}[p_0(x)]$. No independence or convexity hypothesis is used.

Corollary 4 (Pooled rates are conservative)**.** *Assume in addition that $p(x) \equiv p$ is constant. Then h depends on x only through $\varepsilon(x)$ and is convex in it, so $\mathbb{E}[h] \geq h(\bar{\varepsilon})$ and*

$$A_\infty - \bar{p}_0 \;\geq\; \widehat{G} \;+\; \mathrm{Cov}\big(1 - p_0(x), h(x)\big),$$

where $\widehat{G} = (1 - \bar{p}_0)(1 - \bar{\varepsilon})\bar{p}/\bar{q}$ is Cor. 2 evaluated at pooled rates.

The constancy of p is required, not cosmetic: h is convex in ε at fixed p but not jointly, and Appendix E exhibits a population with varying p in which the inequality reverses. Two consequences for §6 are drawn in Appendix D: pooled rates *under*-predict the realised gain, and the covariance is positive exactly when the verifier is more specific on the instances the base model gets wrong.

5 Sound Verifier Stacks

Prop. 2 says the way to make a sound verifier useful is to conjoin more of them. We instantiate stacks on three structured-generation tasks chosen to span a range of expected ε, from a very loose constraint to a tight one.

Table 4 in Appendix G separates candidate checks by whether the soundness argument survives scrutiny, and the separation is not cosmetic. Three checks in common use fail it: a gold SQL query may legitimately return no rows, so a non-empty-result check has $\delta > 0$; a gold query need not name every entity the question mentions; and span-containment is sound for extractive information extraction but not for abstractive. Concatenation is sound for surface segmentation of concatenative morphology and not beyond it. By Cor. 1 these checks remain perfectly usable as the scorer S; they simply must not enter the gate. The three tasks were chosen to span a range of expected ε: We expect SQL executability to be among the loosest nominally sound constraints and surface-form concatenation to be among the tightest, which would make the pair a useful contrast, but both are expectations, not facts we establish here, and Experiment 2 measures them. **[MISSING: decide between morphological segmentation and Universal Dependencies parsing, then fix the checks in Table 4 and Appendix G accordingly]**

Tightness and soundness pull against each other. Assembling Table 4 produced the paper's most useful negative finding, and we report it because it qualifies Prop. 2 instead of illustrating it. The checks one reaches for in order to drive ε down are disproportionately the ones whose soundness argument fails. A non-empty result set is a strong filter on wrong SQL, and it is not sound: a gold query answering a question whose true answer is "none" returns zero rows. Requiring that every entity named in the question appear in the query is a strong filter, and it is not sound: a gold query may reach an entity through a join key without naming it. A segment count inside the attested range is a strong filter, and it is not sound, because it encodes a distributional prior rather than a definitional constraint, and rare but legitimate forms sit outside it. In each case the property that makes a check discriminative, namely that it rules out much of the output space, is the same property that makes it liable to rule out a legitimate gold output.

Prop. 2 is therefore not a licence to accumulate checks. It licenses accumulating checks *whose soundness argument has been made and verified*, and §6 treats every such argument as a hypothesis to be measured rather than a premise. The framework does say what to do with the discriminative unsound checks: Cor. 1 places them in the scorer, where their δ costs nothing because the gate has already run.

Where the leaks are. Each check above is nominally sound, and each has an implementation that can break it. Extraction is the largest surface: a correct query wrapped in prose, fenced with the wrong language tag, or trailing a comment can fail to be recovered, and the check then rejects a correct output. Normalisation is the second: an equivalent query using a different join order, an equivalent segmentation using a different tag alias, an equivalent JSON with reordered keys. A 38.5% false-discovery rate of rejection has been reported in a neighbouring domain, attributed to formatting, notation and equivalence mismatches [6]; those categories overlap ours without coinciding with them. We therefore measure and report δ per check and per language instead of asserting it is zero, and treat a nonzero measurement as a finding about the implementation rather than a nuisance to tune away.

Cost. Every check above is $O(|y|)$ or a single database round trip, and all are computed after generation, not during it, so none constrains the decoder or requires logit access.

6 Experiments

The experiments are ordered so that the cheapest one can refute the account. Experiment 2 is the mechanism gate: it measures the quantities the theory reasons about, and Experiment 3 checks the theory's prediction against the observed effect. If Experiment 3 fails, the decomposition is wrong and the paper reports that, not the method.

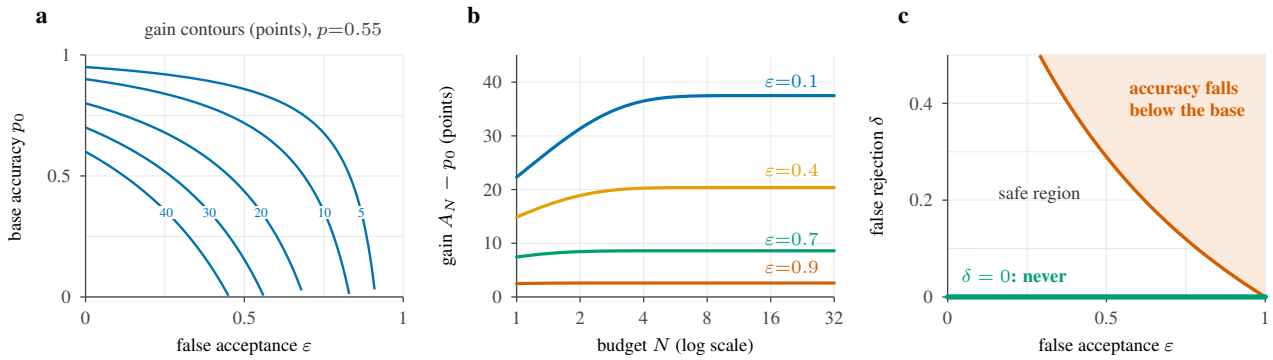


Figure 2: The operating regime, computed from Cor. 2 and Cor. 3. Every curve on this page is a formula evaluated exactly; none is measured. (a) Contours of equal asymptotic gain over the plane of false acceptance and base accuracy. The contours run steeply, so a fixed gain demands a rapidly tighter verifier as the base model improves: the same check that is worth 0.2 against a weak base is worth almost nothing against a strong one, which is why the value of cheap structural checks should fall as models improve. **(b)** Approach to the asymptote with budget. A tight verifier reaches most of its gain within a handful of samples; a loose one has little to converge to, so the familiar shape of a test-time-scaling curve is set by ε rather than by compute. **(c)** Where an unsound verifier costs accuracy, from the exact condition $p_0\delta(1-r) > (1-p_0)(1-\varepsilon)r$ at $p_0 = 0.75$, $p = 0.55$. Above the curve the rule returns worse answers than the base it started from. The green line is the sound edge, which no amount of false acceptance can push into the shaded region: that is the entire content of the guarantee.

Models and data. **[MISSING: fix the model set (at least two families and two sizes, to separate a model effect from a task effect), the exact benchmark versions and splits, the decoding temperature and sampling protocol, and the licences for each dataset] [MISSING: compute environment: hardware, framework versions, and total inference cost]**

Metrics, and one trap. Task accuracy per benchmark. For text-to-SQL we use test-suite accuracy [15] rather than single-database execution match, and the reason is not cosmetic. Execution match labels a query that fails to run as incorrect, so if correctness is defined by execution then the executability check has $\delta = 0$ *by construction* and the soundness audit is vacuous for the one check the paper most needs to audit. Any correctness label that presupposes a check cannot be used to measure that check's error rate. Test-suite accuracy breaks the circularity because it compares behaviour across distilled databases and assigns a label to queries independently of whether one particular execution succeeded. **[MISSING: state the equivalent circularity check for the IE and morphology metrics, and confirm neither correctness label presupposes a check in its own stack] [MISSING: metric for the IE task] [MISSING: metric for the morphology task]**; the measured $p_0, p, \varepsilon, \delta$ per verifier; and mean generated candidates per instance as the compute measure. Every comparison is reported against realised compute, not against the nominal budget N. The gated rule returns at the gate whenever the base is accepted, so at a nominal N it draws far fewer candidates than an ungated rule at the same N; plotting both at equal N would flatter it. We therefore report accuracy against mean generations per instance, and where a single number is quoted it is at matched mean generations. At least five seeds, mean and standard deviation reported, paired bootstrap confidence intervals over instances [3] and a paired permutation test against the strongest baseline, with effect size and n stated alongside every p-value. One point deserves emphasis because the paper's central prediction is a near-null. We predict cheap sound checks buy almost nothing, and a non-significant difference from an underpowered comparison is not evidence for that; it is absence of evidence. We therefore prespecify an equivalence margin and test the near-null claims by two one-sided tests, and report the minimum detectable effect at the achieved sample size for every comparison in which we assert that a gain is small. **[MISSING: fix the equivalence margin before looking at results, state it here, and justify it in terms of what gain would be practically meaningful]**

Baselines. Three groups, and the first is the one that matters.

Rules that differ from ours only in the gate. Ungated best-of-N over the pool including the base, with the same verifier and the same scorer. Homogeneous rejection sampling, which draws until acceptance without treating the base specially and which Dorner et al. [2] show dominates best-of-N at fixed compute. These two isolate the gate, which is the paper's only algorithmic claim, and no result is interpretable without them.

Current selection methods. Execution-clustered self-consistency, which groups candidates by execution result and returns a member of the largest cluster; this is the standard for SQL and string-level voting [14] would be a straw man. A learned outcome reward model [13]. An LLM-as-judge scorer. Li et al. [4], the strongest training-free selection method we are aware of for this task, which cross-checks candidate execution against a parallel imperative implementation on a constructed database. **[MISSING: add the current strongest *trained* selector for each benchmark; check the BIRD and Spider leaderboards at submission time, and state the leaderboard access date]**

Reference points. Greedy decoding, grammar-constrained decoding, and the Experiment 1 oracle.

Experiment 1: headroom (positive control, run first). »> TO RUN: Oracle selection over the candidate pool using gold labels, which are used here and in evaluation only. Output: oracle accuracy minus p_0, per task and model. Gate: if the oracle barely exceeds the base, no verifier can help and the direction is refuted cheaply.

Experiment 2: the soundness audit (mechanism gate, run second). »> TO RUN: Estimate p_0, p, δ and ε directly for every check in §5 and every baseline scorer, per task, per model and per language, with bootstrap intervals. This is the audit that separates nominal from realised soundness, and it is the experiment on which every theoretical result depends. For each nonzero measured δ, classify the cause as extraction, normalisation, or a genuine gap in the nominal argument, and report the breakdown; the first two are implementation defects that can be repaired and re-measured, the third means the check was never sound. Also report the false-discovery rate of rejection alongside δ, so the numbers are comparable with prior reports [6] without converting between them. Output: Tables 1 and the ε column that decides the paper. Gate: if ε is low for the cheap checks, the diagnosis in §7 is wrong and must be reported as such.

Experiment 3: predicted versus observed gain. »> TO RUN: Substitute the Experiment 2 estimates into Cor. 2 and compare the predicted A_N curve against the measured one across $N \in \{1, 2, 4, 8, 16, 32\}$. Report the discrepancy with intervals, not a correlation, and measure the whole curve instead of fitting two points, since Dorner et al. [2] show low-compute behaviour need not extrapolate. If prediction and observation diverge beyond the intervals, report the decomposition as insufficient and diagnose which assumption fails, the leading candidate being the constant-rate idealisation of §3.

Experiment 3b: is it the gate? »> TO RUN: Hold the verifier, the scorer and the candidate pool fixed and vary only the rule: ungated best-of-N, homogeneous rejection sampling, and reject-gated selection. This is the ablation for the paper's only algorithmic claim and it must be run at matched mean generations. Prediction: gated $\geq$ rejection sampling $\geq$ ungated best-of-N in worst-case-over-tasks terms, with the gated-versus-rejection-sampling gap tracking $p_0 - p$. Then sweep decoding temperature so that $p_0 - p$ varies from near zero to large. Prediction, and the sharpest one in the paper: as temperature falls and sampled candidates approach the greedy output, $p_0 - p \to 0$ and the gate's advantage over homogeneous rejection sampling vanishes. If the advantage persists at $p_0 \approx p$, it is not coming from the mechanism we claim and the explanation is wrong. If no configuration gives $p_0 > p$ on these tasks, the gate has no room to help and we report that.

Experiment 4: stacking. »> TO RUN: Add sound checks one at a time in the order given in §5, measuring ε and realised gain after each. Prediction from Prop. 2: ε is non-increasing and gain is non-decreasing, monotonically, at every step. A non-monotone step falsifies either the soundness of that check or the constancy assumption on ε.

Experiment 5: degradation and safe composition. »> TO RUN: Measure δ for the unsound scorers (log-probability, self-consistency, LLM-judge, reward model), evaluate the Cor. 3 condition, and check whether the scorers predicted to degrade actually fall below greedy under unguarded best-of-N. Then run each behind the sound gate and confirm the floor holds, per Cor. 1. This is the experiment that turns the theory into a recommendation, so report the per-task worst case, not an average. Output: Table 2.

Experiment 6: negative control. »> TO RUN: Two deliberately uninformative verifiers. First, a sound check that everything satisfies (the output is a non-empty string), for which the theory predicts exactly zero gain at every N: any measured gain indicates a confound in the pipeline. Second, a scorer with deliberately inflated δ (reject the highest-log-probability

candidate), for which the theory predicts degradation at a rate Cor. 3 specifies in advance. Both are falsifiable and neither is expected to look good. A seventh experiment, repairing the implementation defects Experiment 2 uncovers and re-measuring, is deferred to Appendix C.

Contamination. **[MISSING: report a contamination check for every benchmark against every model's disclosed training data, or state that disclosure is unavailable and treat the affected numbers as an upper bound. Spider and BIRD are old enough and public enough that this cannot be skipped, and p_0 is exactly the quantity contamination inflates]**

7 Predictions

No experiment has been run. This section states what the account predicts, so that it can be checked against measurement rather than written after it. **P1.** Measured ε for parse-and-execute checks on text-to-SQL will be high, which by Cor. 2 would make execution-based best-of-N nearly inert and would account for the ordering Tritto et al. [13] report without appealing to any property of reward models beyond a lower ε. **P2.** At least one check we argue is nominally sound will show a nonzero measured δ, dominated by extraction and normalisation, not by a gap in the nominal argument. We would be surprised if all of them audited clean, given what has been found in a neighbouring domain [6]. **P3.** Stacking the checks in the sound column of Table 4 lowers ε and raises the gain monotonically. We expect the effect to be small, because §5 argues that the discriminative checks are disproportionately the unsound ones and those are excluded from the gate; a large effect here would be welcome and would mean the sound column is tighter than we think. We expect morphological segmentation to show the largest sound-verifier gain of the three tasks, on the expectation that surface-form concatenation is the tightest sound check we have, which Experiment 2 tests. **P4.** At least one unsound scorer falls below greedy on at least one task, and the same scorer behind the sound gate does not.

Each prediction can fail independently. If P1 fails the central diagnosis is wrong, and we will report that the decomposition holds while the empirical premise motivating it does not. If P2 fails, the nominal-versus-realised distinction is real but not urgent, and §5 should be shortened accordingly. If P3 fails in the direction of no effect, Prop. 2 is sound but empty on these tasks and the paper should say so; if P4 fails, Cor. 1 is a guarantee against a danger that does not arise here, which is worth reporting because it is what a practitioner would want to know. **[RP: observed values, with the discrepancy between predicted and measured gain and its confidence interval]**

Limitations

Scope of the theory. It does not predict ε for any verifier, which is an empirical property of the model's error distribution. It does not recover the general ROC characterisation of Dorner et al. [2], which is the right tool away from the top edge, nor their finding that low-compute behaviour need not extrapolate; that finding applies here and is why Experiment 3 measures the whole budget curve instead of extrapolating from its low end. It does not cover adaptive budgets in which N depends on observed scores. A2 assumes verdicts are conditionally independent given correctness, which fails for a stack whose conjuncts share a parse; Prop. 1 and Prop. 2 are unaffected, but Theorem 1 is not, and we do not know the size of that error. **[MISSING: quantify the A2 violation empirically in Experiment 2 by measuring pairwise verdict correlation within each stack]**

The theory is elementary: four short arguments about conditional probabilities, not a deep result, and the general characterisation of verifier imperfection already exists in Dorner et al. [2]. What we add is the behaviour of one distinguished point under one gated rule, the closure property, the composition corollary, and the observation that the guarantee is routinely forfeited by implementations. Reviewers should weigh the paper on those, not on technical difficulty.

The constancy assumption on δ and ε across instances is false in practice, since both vary with difficulty; Prop. 3 removes it exactly, at the cost of a covariance term that must itself be measured. The analysis covers a fixed budget with i.i.d. sampling and does not cover adaptive or sequential allocation, which is how a deployed system would spend compute. Soundness is argued rather than measured, and an argument can be wrong: a check believed sound but actually violated by some gold outputs has $\delta > 0$ and forfeits the floor, which is why Experiment 2 measures δ instead of assuming it. The floor is a floor on expected accuracy over the evaluation distribution and says nothing about any individual instance. Finally, all three tasks are structured-generation tasks in which the notion of a well-formed output is unusually crisp; we make no claim about open-ended generation, where no sound verifier is available and the entire analysis is silent.

Ethics Statement

The work uses public benchmarks and introduces no new data collection or human subjects. The floor guarantee of Prop. 1 concerns expected accuracy and should not be read as a safety property: a system that never degrades on average can still fail badly on individual inputs, and the method must not be presented to downstream users as a correctness guarantee. Sound checks encode assumptions about what a well-formed output looks like, which for the multilingual task risks encoding the conventions of well-resourced languages and penalising legitimate analyses in others; Experiment 2 reports δ per language for exactly this reason, and a nonzero δ concentrated in particular languages is a fairness finding, not a bug to tune away. AI tools assisted

		Measured rates				Gain over base (points)		
Task	Verifier	p_0	p	δ[a]	ε[b]	predicted[c]	observed	Sound?[d]
SQL	parse only	[RP:]	[RP:]	[RP:]	[RP:]	[RP:]	[RP:]	[RP:]
	+ schema valid	[RP:]	[RP:]	[RP:]	[RP:]	[RP:]	[RP:]	[RP:]
	+ executes	[RP:]	[RP:]	[RP:]	[RP:]	[RP:]	[RP:]	[RP:]
	full sound stack	[RP:]	[RP:]	[RP:]	[RP:]	[RP:]	[RP:]	[RP:]
	reward model	[RP:]	[RP:]	[RP:]	[RP:]	[RP:]	[RP:]	*no*
IE	full sound stack	[RP:]	[RP:]	[RP:]	[RP:]	[RP:]	[RP:]	[RP:]
Morph.	full sound stack	[RP:]	[RP:]	[RP:]	[RP:]	[RP:]	[RP:]	[RP:]

[a] Realised false-rejection rate, $\Pr[V{=}0 \mid C{=}1]$. Report the bootstrap interval, not a point estimate: the claim of interest is whether the interval excludes zero.
[b] Realised false-acceptance rate, $\Pr[V{=}1 \mid C{=}0]$. This column decides the paper.
[c] From Cor. 2 evaluated at the measured p_0, p, ε in the same row. Cor. 4 shows this is a *lower* estimate when ε varies across instances, so an observed value above prediction is expected and a value below it is not.
[d] Whether the row's soundness argument survives measurement. A row with δ significantly above zero is not sound as built, whatever the argument says, and by Cor. 1 belongs in the scorer rather than the gate.

Table 1: Verifier rates and the gain they predict (Experiments 2 and 3). Every entry is an unmeasured placeholder. The two right-hand columns are the paper's central test: the theory fixes the predicted column before the observed column is filled, so the comparison is a prediction rather than a fit.

drafting and the derivations; the human authors verified every proposition independently and take full responsibility for all content.

8 Conclusion

Verifier-based test-time scaling has been discussed as though verifier quality were one number. It is two. Soundness controls false rejection and delivers an unconditional floor. It appears in none of the three factors of the gain, which are governed instead by how often the base is wrong, how often the verifier catches it, and how good the replacement is. Specificity enters two of those three, so it is the verifier-side lever, though it cannot manufacture a gain when the model never samples a correct candidate. Structural checks in NLP are nominally sound and, we expect, unspecific, which would make them safe and nearly inert on their own and would explain why learned verifiers that trade a little soundness for a lot of specificity beat them. Whether that expectation survives measurement is the open question, and whether these checks are *realised* sound is the other. The design consequence does not depend on either answer: gate one behind the other.

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Rule	Acc. (%)	Δ vs. greedy[a] (pts, 95 % CI)	Gen.[b] $\bar{n}$	Floor[c]
Reference				
Greedy	[RP:]	—	1.0	—
Constrained decoding	[RP:]	[RP:]	1.0	—
Ungated selection				
Self-consistency[d]	[RP:]	[RP:]	[RP:]	[RP:]
BoN, sound scorer	[RP:]	[RP:]	[RP:]	[RP:]
BoN, reward model	[RP:]	[RP:]	[RP:]	[RP:]
BoN, LM judge	[RP:]	[RP:]	[RP:]	[RP:]
Rejection sampling[e]	[RP:]	[RP:]	[RP:]	[RP:]
Reject-gated (ours)				
Sound stack	[RP:]	[RP:]	[RP:]	**yes**
Sound stack + RM	[RP:]	[RP:]	[RP:]	**yes**
Upper bound				
Oracle selection	[RP:]	[RP:]	[RP:]	—

[a] Paired bootstrap over instances.
[b] Mean generations per instance, the realised compute. Rows are compared at matched $\bar{n}$, not at matched nominal budget: the gated rule returns at the gate whenever the base is accepted, so an equal-N comparison would overstate it.
[c] Whether accuracy stayed at or above greedy on *every* task, not on average. Prop. 1 guarantees this for the gated rows and for no other row here.
[d] Clustered by execution result, not by output string.
[e] Homogeneous: no distinguished first draw. The contrast with the gated rows isolates the gate, the only algorithmic claim in the paper.

Table 2: Main comparison on **[MISSING: primary benchmark]**; remaining tasks in the appendix. Every entry is an unmeasured placeholder.

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A Proofs

Throughout, $C(y) \in \{0,1\}$ is correctness against the gold answer and $V(y) \in \{0,1\}$ is the verifier verdict, as in §3. All statements refer to Definition 2.

Proposition 1 (instance-wise floor). Fix x; every probability below is conditional on x. Realised soundness at x says $C(y) = 1 \Rightarrow V(y) = 1$, whose contrapositive is $V(y_0) = 0 \Rightarrow C(y_0) = 0$. Let $E = \{C(y_0) = 1\}$. On E we have $V(y_0) = 1$, so the first step of Definition 2 returns $\hat{y} = y_0$ and therefore $C(\hat{y}) = 1$. Hence $E \subseteq \{C(\hat{y}) = 1\}$ and

$$\Pr[C(\hat{y}) = 1 \mid x] \;\geq\; \Pr[E \mid x] \;=\; \Pr[C(y_0) = 1 \mid x].$$

The argument never refers to N, to S, to ε, or to the law of $y_1, \ldots, y_N$, so no part of Assumption 1 is used. Taking expectations over x and applying the tower rule gives $\Pr[C(\hat{y}) = 1] \geq \mathbb{E}[\Pr[C(y_0) = 1 \mid x]] = p_0$. □

Corollary 1 (safe composition). The proof above never evaluates S: the scoring function is consulted only inside the sampling branch, which is reached only when $C(y_0) = 0$. The bound is therefore uniform over S. □

Proposition 2 (stacking). *Soundness.* Let $C(y) = 1$. Realised soundness of each V_i gives $V_i(y) = 1$ for every i, hence $V(y) = \bigwedge_i V_i(y) = 1$, so $\delta_V = 0$. *False acceptance.* Fix j. As events, $\{V = 1\} = \bigcap_i \{V_i = 1\} \subseteq \{V_j = 1\}$. Conditioning on $\{C = 0\}$ and using monotonicity of probability,

$$\varepsilon_V = \Pr[V{=}1 \mid C{=}0] \;\leq\; \Pr[V_j{=}1 \mid C{=}0] = \varepsilon_j.$$

Since j was arbitrary, $\varepsilon_V \leq \min_j \varepsilon_j$. Applying the same bound to the conjunction of the first k conjuncts shows ε is non-increasing as conjuncts are added. No independence is used anywhere; the stronger identity $\varepsilon_V = \prod_i \varepsilon_i$ would additionally require the conjuncts to be conditionally independent given $C = 0$, which we do not assume and do not report. □

Theorem 1 (accuracy). Assume A1–A3 and constant S, so that the $\arg\max$ in Definition 2 is a uniform draw from $\mathcal{A}$. Partition on the gate.

(i) Gate passes, base correct. $\Pr[C(y_0){=}1, V(y_0){=}1] = p_0(1-\delta)$ by A2. The rule returns y_0, which is correct. Contribution $p_0(1-\delta)$.

(ii) Gate passes, base incorrect. $\Pr[C(y_0){=}0, V(y_0){=}1] = (1-p_0)\varepsilon$. The rule returns y_0, which is incorrect. Contribution 0.

(iii) Gate fails. $\beta := \Pr[V(y_0){=}0] = p_0\delta + (1{-}p_0)(1{-}\varepsilon)$, and within this event $\Pr[C(y_0){=}1 \mid V(y_0){=}0] = p_0\delta/\beta$. By A1 the N candidates are independent of y_0, and by A2 each is accepted with probability $q = p(1{-}\delta) + (1{-}p)\varepsilon$ independently, so $\Pr[\mathcal{A} = \emptyset] = (1-q)^N$. If $\mathcal{A} \neq \emptyset$, Lemma 1 gives that the returned candidate is correct with probability $r = p(1{-}\delta)/q$, contributing $\beta\big(1{-}(1{-}q)^N\big)r$. If $\mathcal{A} = \emptyset$ the rule returns y_0, correct with probability $p_0\delta/\beta$, contributing $\beta(1-q)^N \cdot p_0\delta/\beta = p_0\delta(1-q)^N$.

Summing (i)–(iii) gives the stated expression. Setting $N = 0$ yields $p_0(1-\delta) + p_0\delta = p_0$, as required. □

Corollary 2 (gain decomposition). Put $\delta = 0$ in Theorem 1. Then $q = p + (1-p)\varepsilon$, $r = p/q$, the term $p_0\delta(1-q)^N$ vanishes, and $\beta = (1-p_0)(1-\varepsilon)$, giving the displayed form. *Monotone in N:* $1-(1-q)^N$ is non-decreasing in N because $0 \le 1-q \le 1$, and every other factor is a non-negative constant. *Monotone in ε:* write $A_\infty - p_0 = (1-p_0)h(\varepsilon)$ with $h(\varepsilon) = (1-\varepsilon)p/\big(p+(1-p)\varepsilon\big)$. Differentiating and using $p + (1-p) = 1$,

$$h'(\varepsilon) = \frac{-p\big(p+(1-p)\varepsilon\big) - p(1-\varepsilon)(1-p)}{\big(p+(1-p)\varepsilon\big)^2} = \frac{-p}{\big(p+(1-p)\varepsilon\big)^2} \;<\; 0$$

□

Corollary 3 (degradation). Let $N \to \infty$ in Theorem 1. If $q > 0$ then $(1-q)^N \to 0$ and $A_\infty = p_0(1-\delta) + \beta r$. Hence

$$A_\infty - p_0 = -p_0\delta + \big[p_0\delta + (1-p_0)(1-\varepsilon)\big]r = -p_0\delta(1-r) + (1-p_0)(1-\varepsilon)r,$$

and $A_\infty < p_0$ iff $p_0\delta(1-r) > (1-p_0)(1-\varepsilon)r$. If $q = 0$ then no candidate is ever accepted, the rule always returns y_0, and $A_\infty = p_0$. □

Proposition 3 (heterogeneous gain). Assume A1–A2 conditionally on each instance with rates $p_0(x), p(x), \varepsilon(x)$ and $\delta(x) = 0$, and drop A3. Corollary 2 applied conditionally on x at $N = \infty$ gives

$$\Pr[C(\hat{y})=1 \mid x] = p_0(x) + \big(1-p_0(x)\big)h(x), \qquad h(x) = \tfrac{(1-\varepsilon(x))p(x)}{q(x)}.$$

Taking expectations and using $\mathbb{E}[UW] = \mathbb{E}[U]\mathbb{E}[W] + \mathrm{Cov}(U,W)$ with $U = 1-p_0(x)$ and $W = h(x)$,

$$A_\infty = \bar{p}_0 + (1-\bar{p}_0)\mathbb{E}[h] + \mathrm{Cov}\big(1-p_0(x), h(x)\big).$$

Only the covariance identity is used, so no independence or convexity hypothesis enters. □

Corollary 4 (pooled rates are conservative). Assume in addition $p(x) \equiv p \in (0,1)$, so $h(x) = h(\varepsilon(x))$ with h as in Corollary 2. Differentiating h' above,

$$h''(\varepsilon) = \frac{2p(1-p)}{\big(p+(1-p)\varepsilon\big)^3} > 0,$$

so h is strictly convex on $[0,1]$ and Jensen's inequality gives $\mathbb{E}[h(\varepsilon)] \ge h(\bar{\varepsilon})$. Since $\widehat{G} = (1-\bar{p}_0)h(\bar{\varepsilon})$, substituting into Proposition 3 yields the claim. □

Remark (the hypothesis in Corollary 4 is necessary). If p also varies, h is a function of two arguments and is *not* jointly convex, so $\mathbb{E}[h(p,\varepsilon)] \ge h(\bar{p},\bar{\varepsilon})$ may fail and with it the inequality. Appendix E exhibits a heterogeneous population with varying p in which the direction reverses. We therefore state Corollary 4 under constant p rather than patching it.

B Auxiliary lemma

Lemma 1 (Acceptance precision)**.** *Under A1–A2, let $\mathcal{A} = \{i \in [N] : V(y_i) = 1\}$ and let I be uniform on $\mathcal{A}$. Then, with $q = p(1-\delta) + (1-p)\varepsilon$ and $r = p(1-\delta)/q$,*

$$\Pr\big[C(y_I) = 1 \;\big|\; \mathcal{A} \neq \emptyset\big] = r.$$

Proof. Under A1–A2 the pairs $\big(C(y_i), V(y_i)\big)$ are i.i.d. across i, with $\Pr[V{=}1] = q$ and $\Pr[C{=}1, V{=}1] = p(1-\delta)$, hence $\Pr[C{=}1 \mid V{=}1] = r$. Fix a non-empty $A \subseteq [N]$ and condition on $\{\mathcal{A} = A\}$. This event constrains $V(y_i) = 1$ for $i \in A$ and $V(y_i) = 0$ for $i \notin A$, and by independence across i it leaves $\{C(y_i)\}_{i\in A}$ i.i.d. with law $\Pr[C{=}1 \mid V{=}1]$. Therefore $\Pr[C(y_I){=}1 \mid \mathcal{A} = A] = r$ for every such A, and averaging over non-empty A preserves the value. □

C Follow-up experiment

Experiment 7: does repairing the implementation restore the floor? »> TO RUN: For every check with a nonzero measured δ in Experiment 2, repair the extraction and normalisation defects it identified, re-measure δ, and re-run Experiment 5. Prediction: the floor violations disappear in proportion to the recovered δ. This is the experiment that makes the nominal-versus-realised distinction actionable rather than merely cautionary, and it is the one a practitioner would care about most.

D Consequences of heterogeneity

Two consequences follow from Cor. 4. First, plugging pooled rates into Cor. 2 *under*-predicts the realised gain, so a measured gain above prediction is evidence of instance heterogeneity rather than of a broken model, and Experiment 3 should report the direction of any discrepancy before diagnosing it. Second, since h decreases in ε, the covariance term is positive exactly when the verifier is more specific on the instances the base model more often gets wrong. That is a measurable property of a verifier and, we suspect, a better optimisation target than average ε.

Verifier	Reported	Source
Rule-based math checker	38.5% of rejections were correct answers	[6]
LM judges, prefix attack	**[VERIFY: acceptance rate range not confirmed; read it off the paper's own tables or replace this cell with the qualitative finding]**	[7]
Execution vs. reward model, SQL	reward model ahead by up to 4.33 (BIRD), 2.10 (Spider)	[13]

Table 3: Verifier error figures reported by other authors, reproduced to motivate the two-rate view. **These are not our measurements.** Note that the first row is the false-discovery rate of rejection, $\Pr[C{=}1 \mid V{=}0]$, not $\delta = \Pr[V{=}0 \mid C{=}1]$; the two differ by the base rate and we do not convert between them.

E Numerical verification

Every statement in §4 was checked against Monte Carlo simulation of Definition 2. This checks the algebra, not the modelling assumptions, and is not evidence about any language model. Code accompanies the submission.

Statement	Check
Thm. 1	closed form vs. simulation over 8 settings spanning $\delta \in [0, 0.4]$, $\varepsilon \in [0.2, 0.95]$, $N \leq 32$, 3×10^5 trials each: max absolute discrepancy 0.0013
Thm. 1, $N{=}0$	returns p_0 exactly in all settings
Cor. 2	agrees with Thm. 1 at $\delta = 0$ to machine precision over an 81-point grid
Cor. 2 monotonicity	non-decreasing in N and non-increasing in ε on the grid
Cor. 3	predicted degradation matched observed degradation in $4/4$ test settings
h strictly convex	minimum second difference $> 2 \times 10^{-6}$ for $p \in \{0.2, 0.5, 0.8\}$
Prop. 3	exact identity reproduced to $< 10^{-6}$ with p constant and with p varying
Cor. 4	holds in $6/6$ populations with p constant; **reverses** in $2/2$ populations with p varying and ε correlated with p_0, confirming the hypothesis is necessary

F Reported verifier error rates

Task	Nominally sound	*Not* sound: fails on a gold output
Text-to-SQL	parses; schema-valid; executes; operators type-check against column types; every projected non-aggregate appears in GROUP BY	non-empty result; question entities appear in predicates; projected type matches expected answer type
Schema-constrained IE	parses as JSON; validates against schema; required fields present; no schema-forbidden overlap	every span is a literal substring *(sound only if the task is purely extractive)*
Morph. segmentation	segments concatenate to the surface form *(surface segmentation of concatenative morphology only)*; no empty segment; tags in the inventory	segment count in attested range

Table 4: Candidate checks, separated by whether the soundness argument actually holds. Left column: a gold output satisfies the check by definition of the task, so Prop. 2 applies and any subset may be conjoined into the gate. Right column: checks in common use for which we could construct a gold counterexample, so they have $\delta > 0$ and must *not* enter the gate; by Cor. 1 they remain usable as the scorer S. The italicised parentheses mark scope conditions under which a left-column check stops being sound. Full definitions in Appendix G.

```
>> TO RUN: give the exact definition,
tolerance and implementation of
every check in Table 4, including
the extraction and normalisation
steps, since those are where realised
soundness is lost (§5).
```

G Verifier stack definitions